

Relief Forge Tools: Validation

Synthetic test geometry and measured accuracy

Abstract

This document records how the geometry operations in this repository are verified. Every test case is constructed in closed form, so expected values derive from geometry rather than from a previous run of the code under test. All tables are generated from live execution by `docs/make_tables.py` and `scripts/benchmark.py`; none are transcribed by hand.

1 Synthetic test geometry

The test suite uses no binary assets. Every fixture is built analytically at run time, which makes the suite deterministic, fast, and executable without network access, model weights or a GPU. Fixtures fall into three groups.

Valid solids. A closed axis-aligned cube provides the reference case: its volume, surface area, edge count and mean dihedral angle are all known exactly. An annulus — a washer with a through hole — provides a valid solid of genus one, confirming that no check incorrectly assumes a simply connected surface. A subdivided octahedron approximates a sphere and provides a case where the expected value is approached rather than met exactly.

Defect fixtures. Each is the reference cube with exactly one defect introduced, so a detection failure localises immediately: one face removed produces boundary edges; one triangle reversed produces inconsistent winding; a two-triangle flap attached to an existing edge produces a non-manifold edge; two separated cubes produce a second connected component; and reversing every triangle produces negative signed volume while leaving the surface closed.

Symmetry fixtures. A box spanning $x \in [-10, 5]$ exercises an off-centre cut with a known result. A sphere displaced along x exercises a cut whose expected volume follows from the spherical-cap formula. The annulus exercises the case where vertices lie exactly on the cut plane.

2 Analytical expected values

Expected values are closed-form wherever the geometry permits. For the cube of edge a : $V = a^3$, $A = 6a^2$. For the annulus of outer radius R , inner radius r and height h :

$$V = \pi (R^2 - r^2) h, \quad A = 2\pi (R + r) h + 2\pi (R^2 - r^2) \quad (1)$$

For the symmetrized displaced sphere, the retained portion is a spherical cap of height $h = R - d$ on a sphere of radius R displaced by d ; the mirrored result is two such caps:

$$V = 2 \cdot \frac{\pi h^2 (3R - h)}{3} \quad (2)$$

The cube's mean dihedral angle is obtained by counting rather than by formula. A cube tessellated as two triangles per face has 18 unique edges: 12 geometric edges at 90° and 6 face

diagonals at 0° , giving a mean of exactly 60° . This is a strong test, since an error in edge pairing changes the count and therefore the mean.

Table 1 reports measured values against these expectations.

Quantity	Unit	Expected	Measured	Rel. error	Tolerance
Cube volume	mm ³	1 000.0000	1 000.0000	$0.00e + 00$	$1e - 09$
Cube surface area	mm ²	600.0000	600.0000	$0.00e + 00$	$1e - 09$
Cube mean dihedral angle	deg	60.0000	60.0000	$0.00e + 00$	$1e - 09$
Cube cast mass, silver 925	g	10.3600	10.3600	$0.00e + 00$	$1e - 09$
Annulus volume	mm ³	791.6813	790.4102	$1.61e - 03$	$3e - 03$
Annulus surface area	mm ²	791.6813	790.7280	$1.20e - 03$	$3e - 03$
Sphere volume	mm ³	2 144.6606	2 141.5041	$1.47e - 03$	$1e - 02$
Symmetrized sphere volume (cap formula)	mm ³	1 357.1680	1 354.7132	$1.81e - 03$	$2e - 02$
Symmetrized box volume	mm ³	960.0000	960.0000	$0.00e + 00$	$1e - 09$

Table 1: Measured values against analytical expectations. Relative error and the tolerance applied by the test suite are shown for each quantity.

3 Numerical tolerances

Two tolerance regimes apply, and the distinction is deliberate.

Exact cases use a relative tolerance of 10^{-9} . The cube’s volume, surface area and mean dihedral angle, and the symmetrized box’s volume, are representable in double precision without accumulation error; the summations involved are short and well conditioned. A tolerance this tight would fail on any error in the algorithm rather than in floating point.

Discretisation cases use tolerances matched to a known approximation error. The annulus is a 64-segment polygonal approximation of a circle. Its outline is inscribed, so it under-estimates the true area by a factor of $\text{sinc}(1/n) \approx 1 - \pi^2/(6n^2)$, about $8e - 4$ at $n = 64$. The tolerance is set at $3e - 3$, comfortably above that bound but far below any plausible algorithmic error. The test additionally asserts that the measured volume is *less* than the exact value, which verifies the sign of the discretisation error rather than only its magnitude.

The subdivided-octahedron sphere is likewise inscribed and converges more slowly, so it carries a $1e - 2$ tolerance, and the symmetrized sphere $2e - 2$, reflecting the compounding of two approximations.

Stating tolerances by regime rather than using one loose global value means a tightening of the discretisation cannot silently mask a regression in the exact cases.

4 Failure detection

Table 2 records what the analysis reports for each fixture. Each defect fixture must be detected as exactly its own defect, and the valid solids must produce no false positives — including the annulus, whose through hole is valid geometry and must not be reported as a defect.

Two failure modes deserve specific mention because they are easy to miss.

An **inverted mesh** — every triangle reversed — is a closed manifold and passes every edge-incidence test. Only the sign of the signed volume distinguishes it from a valid solid. This is why orientation is checked independently of topology rather than inferred from it.

Disconnected solids are each individually valid. Two separated cubes have no boundary edges,

Fixture	Injected defect	Detected
Closed cube	none	none
Cube, one face removed	open surface	4 boundary
Cube, one triangle reversed	winding	3 winding
Cube with attached flap	non-manifold	3 boundary, 1 non-manifold
Two separated cubes	disconnected	2 components
Annulus (genus one)	none	none
Inverted cube	orientation	negative volume

Table 2: Injected defects and the classification reported by the analysis.

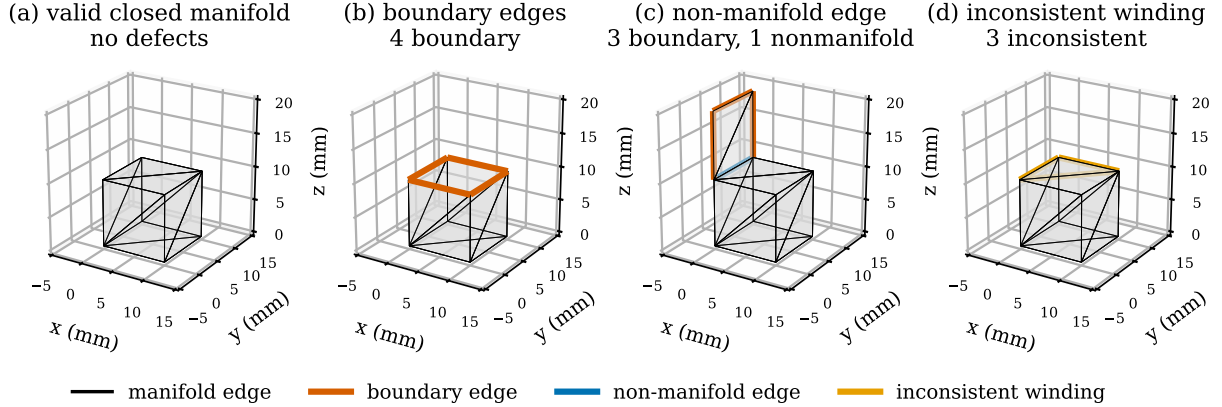


Figure 1: Edge classification on synthetic meshes. Defective edges are drawn in the colour of their classification; all four panels share one camera and one set of axis limits.

no non-manifold edges and consistent winding throughout. Only the component count reveals the condition, which is why the expected component count is an explicit parameter rather than an assumption.

4.1 A defect found by these tests

The plane-intersection cases were added after the symmetry operation produced a non-manifold result on the annulus. Vertices of that fixture lie exactly on the cut plane, and three degenerate cases followed: triangles lying wholly in the plane were duplicated by their own mirror image; triangles on the discarded side touching the plane at one vertex clipped to zero-area slivers; and intersections computed at vertices already on the plane repeated points in the output polygon. The current implementation handles all three, and the annulus case is retained specifically to prevent regression.

5 Performance

Table 3 reports measured timings across synthetic mesh sizes. Values come from actual runs on the machine and software versions recorded in the table’s source file; they are not estimates and are not transferable to other hardware.

Topology analysis dominates, and its cost is superlinear in triangle count because connected-component labelling uses a Python-level union-find loop while parsing and welding are vectorised. This is the obvious optimisation target should larger meshes become routine; it has not been undertaken because meshes at the scale this pipeline produces are analysed in well under a second.

Triangles	File (MB)	Parse (ms)	Weld (ms)	Topology (ms)	Total (ms)	Peak mem. (MB)
512	0.024	0.01	0.40	3.07	3.49	0.21
2 048	0.098	0.02	1.53	12.36	13.91	0.74
8 192	0.391	0.04	6.62	50.66	57.33	2.96
32 768	1.563	0.18	27.51	207.86	235.56	11.82

Table 3: Measured cost of parsing, welding and topology analysis. Median of five runs per configuration.

6 Known limitations

Scope of the checks. A mesh passing validation has no boundary edges, no non-manifold edges, no degenerate edges, consistent local winding, positive signed volume, and the expected component count. This establishes that it is a closed, coherently oriented solid. It does not assess minimum wall thickness, unsupported overhangs, printer tolerances, casting shrinkage, trapped volumes, or any process-specific constraint.

Self-intersection is not detected. A surface may be closed, manifold and correctly wound while passing through itself. Detecting this requires pairwise triangle intersection tests, which are not implemented.

Welding tolerance is heuristic. Vertices closer than the tolerance are merged. On a mesh with genuine detail below that scale, welding could join surfaces that should remain distinct. The default of 10^{-6} of the bounding-box diagonal is far below the resolution of any casting process, but it is a choice rather than a derived bound.

Mean dihedral angle is a limited proxy. It is interpretable only when comparing closely related meshes of the same object at identical scale and comparable triangle density. It is not a perceptual quality metric and carries no meaning across unrelated meshes or across tessellation densities.

Material mass assumes a dense solid. Equation $m = \rho|V|$ excludes sprue, flashing and shrinkage, and alloy densities vary between suppliers. The estimate supports comparison between candidate geometries; it does not certify a finished part.

Generation is not covered by these tests. The test suite validates geometry processing only. Backend inference requires model weights, a GPU and network access, and is therefore excluded from automated testing.